

Model Selection Research Notes

Supporting document for Decision Log 2 - StackEval Benchmark

Project	StackEval Benchmark - Professional Research Project
Decision focus	Selecting the latest LLM models across different categories and ecosystems
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Purpose	To document the reasoning, website review, and evidence behind the updated model selection

Executive summary: The first version of the decision log focused on widely used models. After feedback, extra research, and the separate decision to use API-based querying in Decision Log 3, the selection strategy changed. The final choice prioritizes the latest publicly available models that are accessible through APIs and represent different categories of the current LLM ecosystem.

1. Why this supporting document exists

This document explains the research process behind the updated model selection in Decision Log 2. It is intended as a more detailed companion text for anyone who wants to understand how the final choice was made, which websites were reviewed, what was checked on those websites, and why the final decision differs from the first version.

2. Starting point and change in thinking

The first version of the decision log selected models mainly because they were widely used and often mentioned in benchmark discussions. After receiving feedback and continuing the research, that approach was reconsidered. The key insight was that the LLM ecosystem changes quickly, so a model that is widely used in older benchmark discussions may no longer represent the current state-of-the-art.

A second important change came from Decision Log 3. Once API-based querying was selected as the benchmark execution method, model selection could no longer be treated only as a theoretical comparison exercise. The chosen models also had to be practically usable in an automated setup. That meant I had to verify whether the models were accessible through official or stable API platforms.

3. What I was looking for during the research

- A clear frontier proprietary model
- A competing frontier model from another ecosystem
- An open-source model
- A budget or lightweight model
- The latest publicly available version in each relevant category
- API access or a realistic API path for experimentation
- Enough documentation to justify the decision in a portfolio or decision log

4. Website review and observations

To make the decision stronger, I reviewed the websites and platforms below. I did not use them only as links to cite later. I used them to answer practical research questions such as: Which models are currently presented by the provider? Which names are official? Are they accessible through API documentation? Do benchmark-style platforms already compare them?

Source	What I checked	What I found	Why it mattered
OpenAI models page	I checked the official model documentation to confirm the current OpenAI model names and to see which product/API names were actively presented.	I observed GPT-5.4 as the frontier model and GPT-5.4-mini as the lighter budget option on OpenAI's developer-facing material.	This supported the decision to use the latest OpenAI models instead of older OpenAI baselines.
Anthropic Claude docs	I reviewed the Claude platform documentation to confirm which Claude versions were currently	I observed Claude Sonnet 4.6 as a current documented model choice for the Anthropic ecosystem.	This made it possible to choose a competitor model that was both current and practically usable in an API

	listed and usable through Anthropic's platform.		workflow.
OpenRouter Meta Llama page	I checked how Meta Llama models could be accessed through an API platform and whether the Llama family was represented there as a realistic open-model option.	I observed Llama 4 as an available model family through an API-accessible provider route.	This helped solve the practical issue that open models may not always be used directly through a first-party API in the same way as proprietary models.
ProLLM leaderboard	I reviewed a benchmark-style comparison platform to see whether the selected model families also appear in broader model comparisons.	I observed that some of the selected model families were already being compared in benchmark-style leaderboards.	This added extra confidence that the chosen models are relevant for evaluation and not random picks.

5. How Decision Log 3 influenced this decision

Decision Log 3 concluded that API-based querying is the better method for the StackEval benchmark because it improves consistency, reproducibility, automation, and professional alignment. That decision directly affected model selection. Once the benchmark was going to be executed through APIs, I needed to verify that the selected models were not only interesting from a comparison perspective, but also feasible to query in practice.

Because of this, the final selection became more precise. I was no longer only asking 'Which models represent different categories?' I was also asking 'Which latest models can actually be integrated into the experiment through APIs or stable API platforms?'

6. Why the final selection changed from the first version

First version logic	Updated logic
Focus on widely used or often referenced models	Focus on the latest publicly available models
Main question: which models are common in benchmark discussions?	Main question: which models best represent the current state-of-the-art and can be used through API-based benchmarking?
Selection mainly justified through popularity and benchmark frequency	Selection justified through provider documentation, API accessibility, ecosystem representation, and benchmark relevance

7. Final model selection and rationale

Category	Model	Ecosystem	Why this model fits	API path
Frontier proprietary	GPT-5.4	OpenAI	Represents the latest flagship OpenAI model and therefore a strong frontier baseline for the benchmark.	Direct
Frontier competitor	Claude Sonnet 4.6	Anthropic	Represents a current competing proprietary ecosystem and supports cross-provider comparison.	Direct
Open-source	Llama 4	Meta AI	Represents the open-model side of the ecosystem while still being realistic to access through an API platform.	Via platform
Budget / lightweight	GPT-5.4-mini	OpenAI	Represents a lighter and cheaper model option, enabling comparison between frontier and budget tiers.	Direct

8. Why this final decision is stronger

- It reflects the current state of the ecosystem instead of relying on older benchmark defaults.
- It stays aligned with the API-based execution method chosen in Decision Log 3.
- It compares different categories and ecosystems at the same time.
- It is easier to justify academically because the decision is connected to explicit evidence from provider and benchmark websites.
- It remains manageable: four models are enough to create meaningful comparisons without making the analysis too large.

9. Detailed evidence notes

The evidence below can be used as portfolio support or as explanation material if someone asks how the updated decision was reached.

OpenAI documentation

I used OpenAI's models documentation to verify which GPT models were currently positioned on the developer side. This was important because I needed the product/API naming, not only the names shown in the ChatGPT interface. The page helped support the move from older GPT-4o-style reasoning to the latest GPT-5.4 family selection.

Anthropic documentation

I used Anthropic's Claude platform documentation to confirm that the selected Claude model was not just a random leaderboard name, but part of the current platform and relevant for API-based use.

OpenRouter Meta Llama access page

I used this platform as practical evidence that the Llama family can be included in an API-driven benchmark even when access is mediated through a provider platform. This matters because it makes the open-source category feasible rather than purely theoretical.

ProLLM leaderboard

I used the leaderboard as a supporting benchmark reference to see whether the model families I was considering already appear in comparison-oriented environments. This did not replace official provider documentation, but it strengthened the relevance of the selected families.

ChatGPT Socratic dialogue

I used the dialogue to structure the reasoning, challenge earlier assumptions, and move step by step from a popularity-based selection to a current, API-aware selection strategy.

10. Suggested wording for explaining the decision

After receiving feedback on the first version of the decision log, I reviewed official provider websites and API platforms to verify which models were currently available and usable for an automated benchmark. Combined with the separate decision to use API-based querying in Decision Log 3, this changed my model selection strategy. Instead of selecting the most widely used models, I selected the latest publicly available models across different categories and ecosystems: a frontier model, a competing proprietary model, an open-source model, and a budget model.

11. APA-style source list

Anthropic. (2026). Claude API platform. <https://platform.claude.com/docs/en/home>

OpenAI. (2026). ChatGPT Socratic dialogue discussion [Large language model].
<https://chatgpt.com/share/69b7f668-b368-8002-90b6-d4634e666ecd>

OpenAI. (2026). Models API and platform overview. <https://developers.openai.com/api/docs/models>

OpenRouter. (2026). Meta Llama models via API platform. <https://openrouter.ai/meta-llama>

ProLLM. (2026). LLM leaderboard and benchmark comparisons. <https://www.prollm.ai/leaderboard>